

Setting up a Frontend Web Development Environment

IT 3203

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Overview

- This tutorial will help you set up your own web development environment and begin the fun of web development (for all exercises and projects in this class).
- All information is valid for the semester indicated in the title slide. Things may change in the future and please use it just as a reference.
- **Please complete this within the first two weeks of the semester!**

Development Environment Summary

- GitHub Pages <https://pages.github.com>
 - Web hosting (public web server) that hosts your GitHub website (free)
- Visual Studio Code <https://code.visualstudio.com>
 - Code editor / IDE (integrate development environment)
 - Git is also integrated in VS Code.
- CodePen <https://codepen.io>
 - Online coding playground for quick testing!
- GitHub Desktop <https://github.com/apps/desktop>
 - The GUI tool to manage source code and code publication

All are free!

GitHub Pages

<https://pages.github.com>

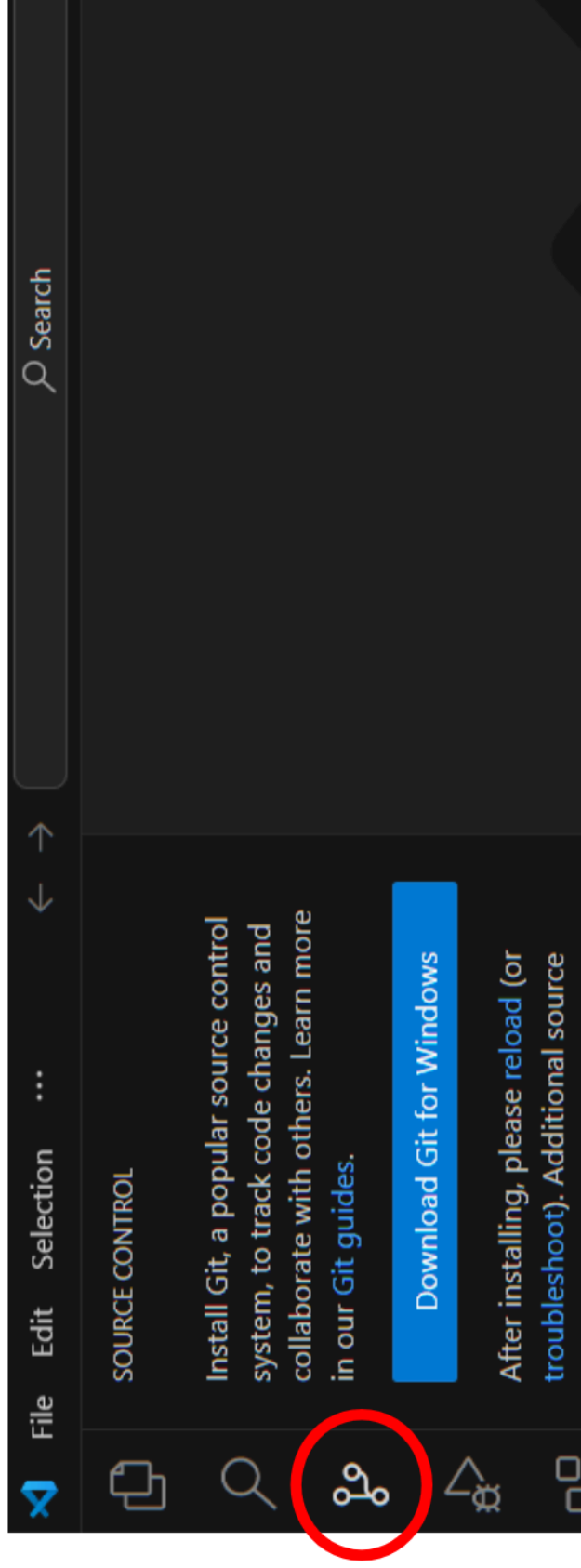
- GitHub pages serves websites directly from GitHub repositories
 - Need a GitHub account
 - The course website is hosted by GitHub Pages <https://ksuweb.github.io/IT3203/>
- Features
 - Provides free services for Git based projects.
 - Only hosts client-based web sites/apps (HTML/CSS/JavaScript)
 - Good for frontend development
- How to set up a root level GitHub website?
 - <https://www.khanacademy.org/computing/computer-programming/html-css/web-development-tools/a/hosting-your-website-on-github>
- Also follow <https://pages.github.com> for quick setup for a project site (subsite in a subdirectory)
 - The guide to set up a website is right on this page. Please follow the steps:
 - First choose “project site” and then “Start from scratch”
 - Just follow the rest of the steps

Manage repositories and source codes updates

1. Use the built-in upload tools on the GitHub website
 - see the Khan Academy tutorial in previous slide
2. Use VS Code integrated GitHub extension – recommended
 - See next slide
- Note in this class, we use GitHub extension more like a file publishing tool, rather than as source and version control tool.

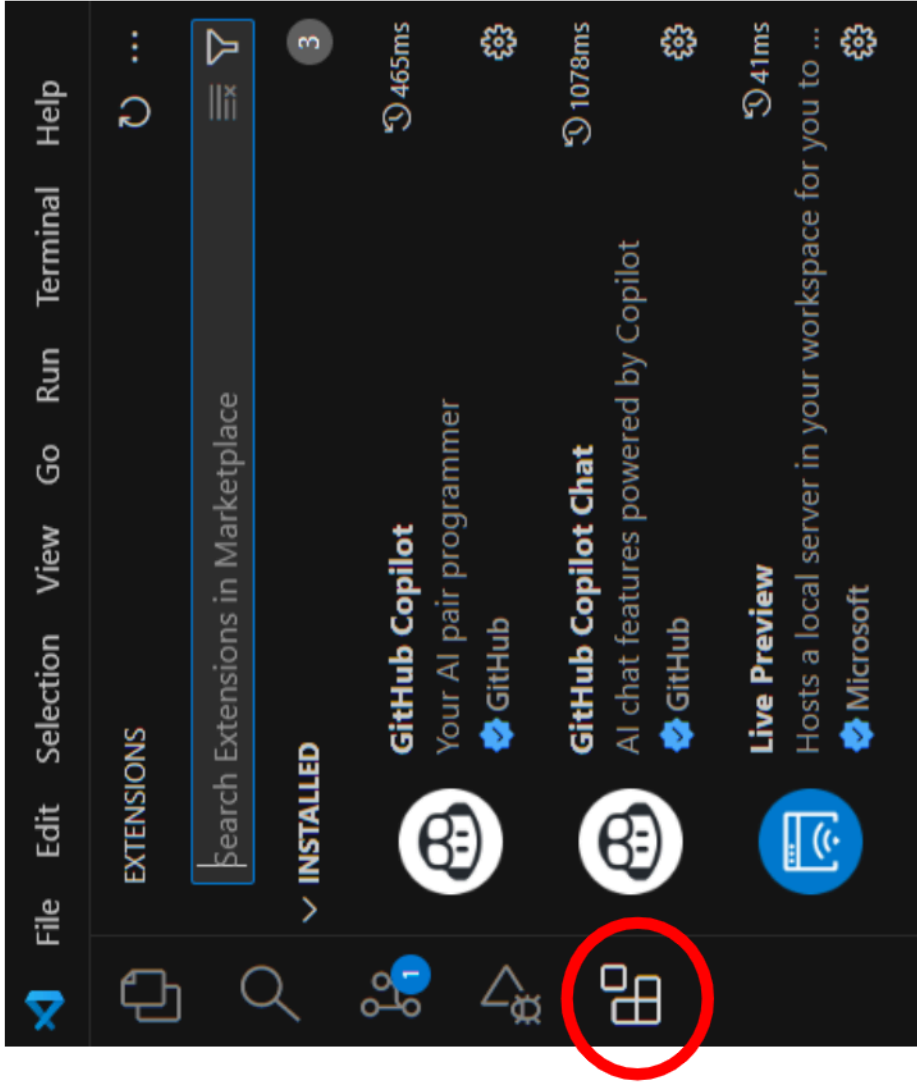
Use Git Directly in VS Code

- Download and install Git system. See a quick setup
 - <https://www.youtube.com/shorts/r-C7s15vDM>
- For more info, visit
 - <https://code.visualstudio.com/docs/sourcecontrol/overview>
 - <https://code.visualstudio.com/docs/sourcecontrol/intro-to-git>



Useful VS Code Extensions

- Recommended
 - Live Preview
 - GitHub Copilot
 - GitHub Copilot Chat
- Go to Extensions and look for these to install



Live Preview

- Live preview allows instant preview while editing



KSU IT 3203 Introduction to Web Development

[Home](#) | [Content](#) | [Website Examples](#)

Course Introduction

This course introduces the basics of web technologies and web development. The course focuses on frontend development using HTML, CSS, and JavaScript.

Tools/services used in this class

[Visual Studio Code](#) | [GitHub Pages](#) | [Code Pen](#)

Course Content and Materials

See [Content Page](#)

Learning Resources

General Learning Resources for Web Development

[W3Schools](#) | [MDN Web Docs](#) | [web.dev](#) | [freeCodeCamp](#) | [SitePoint](#)

Instructor's Learning Materials Related to Web Development

- [Mobile and Web Lecture Notes Hosted on EDOCR](#)
- [Instructor's Examples Hosted on CodePen](#)
- [KSU IT 3203 Intro to Web](#)
- [KSU IT 4213 Mobile Web](#)
- [KSU IT 4203 Advanced Web](#)
- [KSU IT 5443 Web Tech](#)

Tool Alternatives (for reference only)

- Code editor or IDE (integrate development environment)
 - Notepad ++ <https://notepad-plus-plus.org>
 - Honorable mention: <http://brackets.io>
- **GitHub Desktop** to Manage GitHub Repositories
 - [Video] Git, GitHub, & GitHub Desktop for beginners <https://www.youtube.com/watch?v=8Dd7KRpKeaE>
 - How to Use GitHub Desktop: A GitHub Desktop Tutorial <https://www.simplilearn.com/how-to-use-github-desktop-tutorial-article>
- WinSCP <https://winscp.net>
- You may use other tools if you are familiar with them.

All are free!

Other Free Web Hosting that supports server-side development (for reference only)

- Microsoft Azure
 - <https://azureforeducation.microsoft.com/en-us/Institutions>
 - More advanced features and more server-side support
 - Good for advanced web development class
 - Sample Azure hosted sites for other web classes
 - <https://it4203.azurewebsites.net>
 - <https://it4213.azurewebsites.net>
 - <https://it5443.azurewebsites.net>
- Others
 - <https://render.com> – also uses GitHub for repositories
 - <https://glitch.com> – using online code editor